LAPORAN PRAKTIKUM
TEKNOLOGI IOT II

5.7 Tugas

Dosen Pengampu:

1. Dr. Ir. Septafiansyah Dwi Putra - NIP : 198809112009121001
2. Agus Ambarwari, S.Pd., M.Kom. - NIP : 199208082019031020



Disusun oleh :

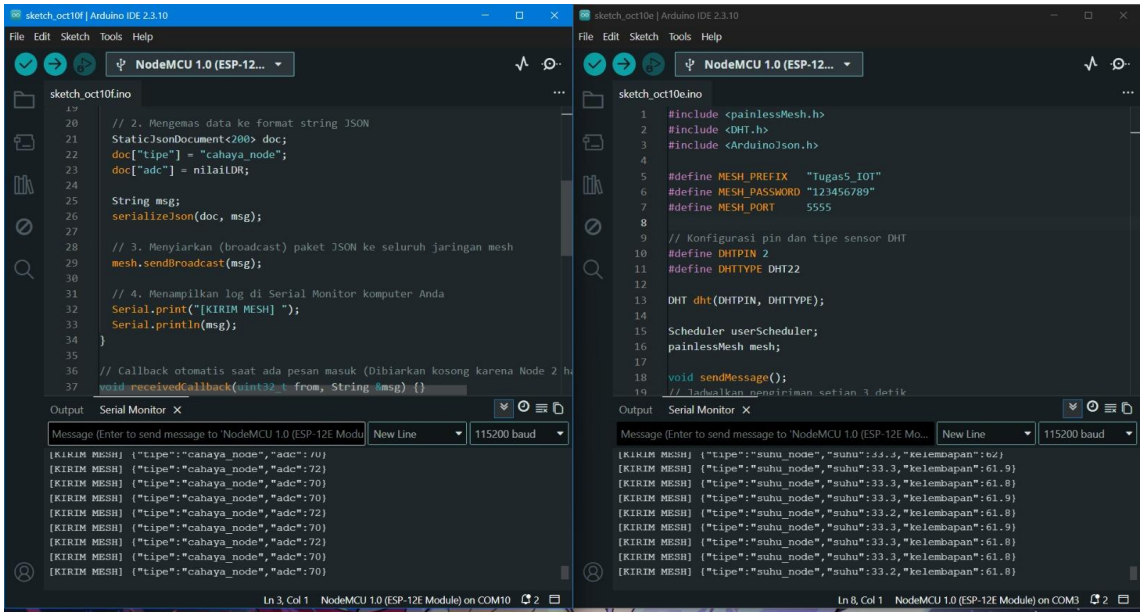
Ananda Haris Rahmadi (25782032)

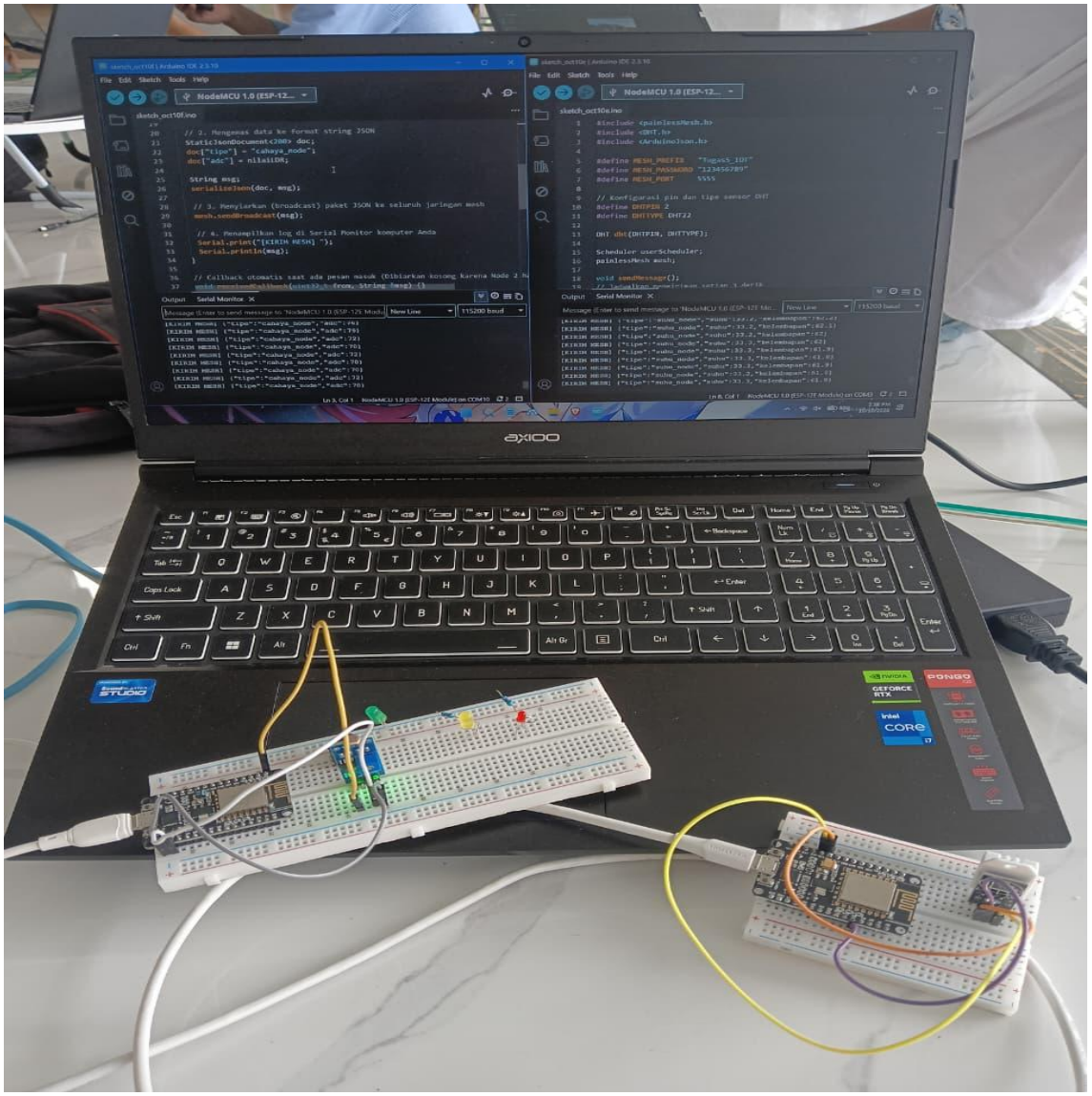
PROGRAM STUDI TEKNOLOGI REKAYASA INTERNET

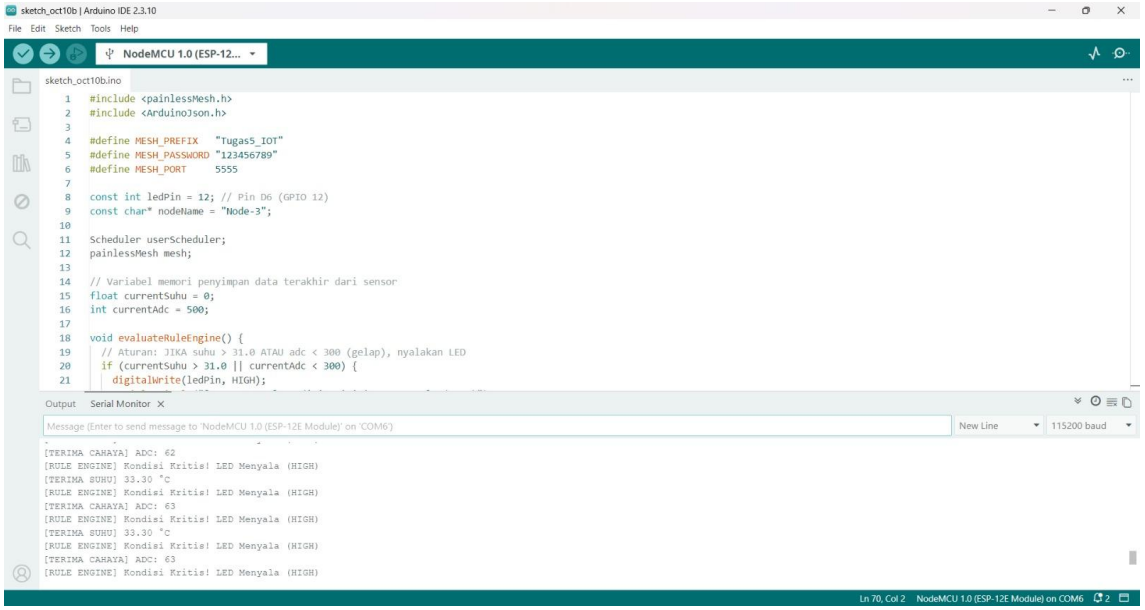
JURUSAN TEKNOLOGI INFORMASI

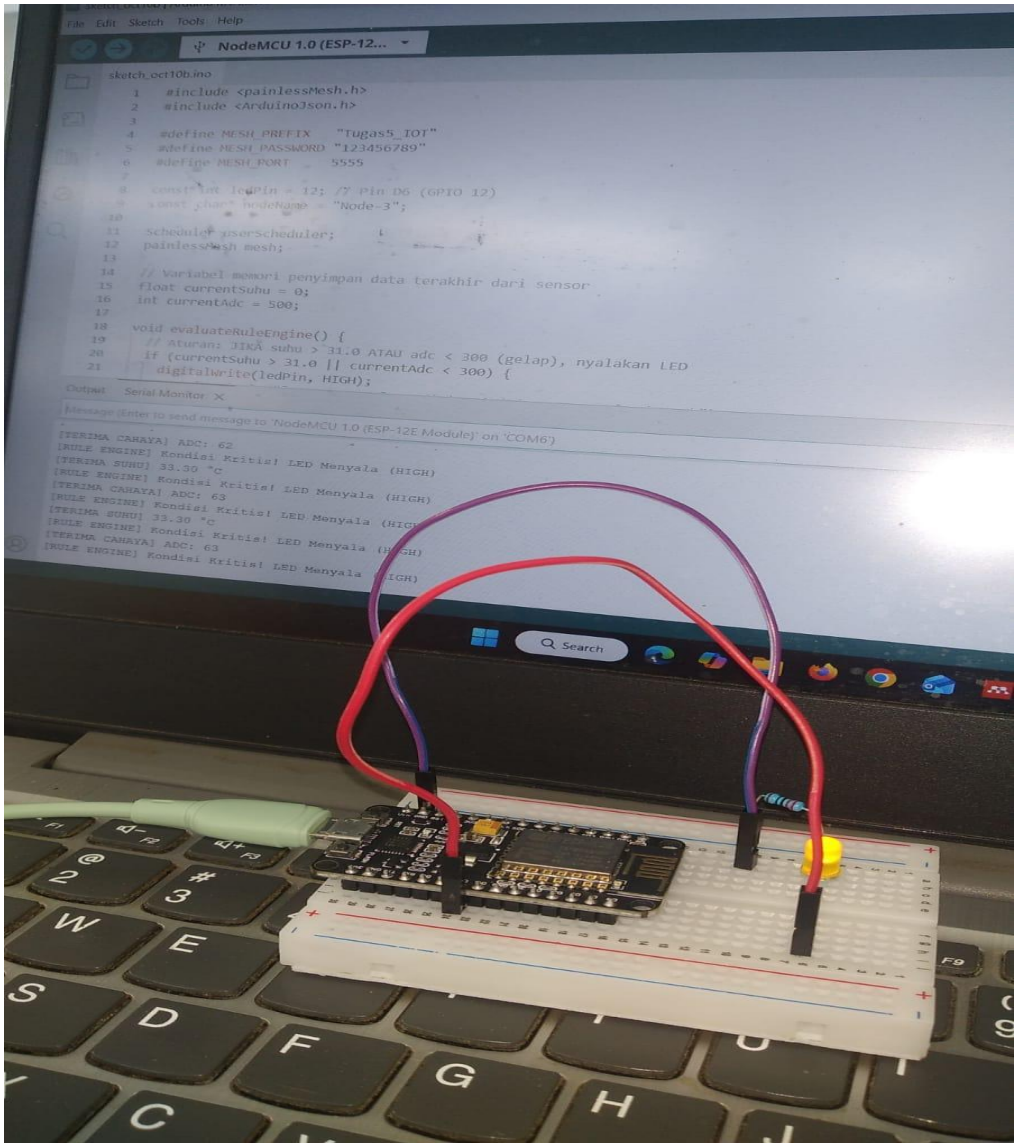
POLITEKNIK NEGERI LAMPUNG

2026


sketch_oct10f | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_oct10f.ino
// 2. Mengemas data ke format string JSON
StaticJsonDocument<200> doc;
doc["tipe"] = "cahaya_node";
doc["adc"] = nilaiLDR;
String msg;
serializeJson(doc, msg);
// 3. Menyiarkan (broadcast) paket JSON ke seluruh jaringan mesh
mesh.sendBroadcast(msg);
// 4. Menampilkan log di Serial Monitor komputer Anda
Serial.print("[KIRIM MESH] ");
Serial.println(msg);
}
// Callback otomatis saat ada pesan masuk (Dibiarkan kosong karena Node 2 h
void receivedCallback(uint32_t from, String &msg) {}
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Modu
New Line
115200 baud
[KIRIM MESH] {"tipe":"cahaya_node","adc":70}
[KIRIM MESH] {"tipe":"cahaya_node","adc":72}
[KIRIM MESH] {"tipe":"cahaya_node","adc":70}
[KIRIM MESH] {"tipe":"cahaya_node","adc":70}
[KIRIM MESH] {"tipe":"cahaya_node","adc":72}
[KIRIM MESH] {"tipe":"cahaya_node","adc":70}
[KIRIM MESH] {"tipe":"cahaya_node","adc":72}
[KIRIM MESH] {"tipe":"cahaya_node","adc":70}
[KIRIM MESH] {"tipe":"cahaya_node","adc":70}
Ln 3, Col 1 NodeMCU 1.0 (ESP-12E Module) on COM10
sketch_oct10e | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_oct10e.ino
#include <painlessMesh.h>
#include <DHT.h>
#include <ArduinoJson.h>
#define MESH_PREFIX "Tugas5_IOT"
#define MESH_PASSWORD "123456789"
#define MESH_PORT 5555
// Konfigurasi pin dan tipe sensor DHT
#define DHTPIN 2
#define DHTTYPE DHT22
DHT dht(DHTPIN, DHTTYPE);
Scheduler userScheduler;
painlessMesh mesh;
void sendMessage();
// Jadwalkan pengiriman setiap 3 detik
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Mo...
New Line
115200 baud
[KIRIM MESH] {"tipe":"suhu_node","suhu":33.3,"kelembapan":61.9}
[KIRIM MESH] {"tipe":"suhu_node","suhu":33.3,"kelembapan":61.8}
[KIRIM MESH] {"tipe":"suhu_node","suhu":33.3,"kelembapan":61.9}
[KIRIM MESH] {"tipe":"suhu_node","suhu":33.2,"kelembapan":61.8}
[KIRIM MESH] {"tipe":"suhu_node","suhu":33.3,"kelembapan":61.9}
[KIRIM MESH] {"tipe":"suhu_node","suhu":33.3,"kelembapan":61.8}
[KIRIM MESH] {"tipe":"suhu_node","suhu":33.3,"kelembapan":61.8}
[KIRIM MESH] {"tipe":"suhu_node","suhu":33.2,"kelembapan":61.8}
Ln 8, Col 1 NodeMCU 1.0 (ESP-12E Module) on COM3




sketch_oct10b | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
sketch_oct10b.ino
#include <painlessMesh.h>
#include <ArduinoJson.h>
#define MESH_PREFIX "Tugas5_IOT"
#define MESH_PASSWORD "123456789"
#define MESH_PORT 5555
const int ledPin = 12; // Pin D6 (GPIO 12)
const char* nodeName = "Node-3";
Scheduler userScheduler;
painlessMesh mesh;
// Variabel memori penyimpan data terakhir dari sensor
float currentSuhu = 0;
int currentAdc = 500;
void evaluateRuleEngine() {
// Aturan: JIKA suhu > 31.0 ATAU adc < 300 (gelap), nyalakan LED
if (currentSuhu > 31.0 || currentAdc < 300) {
digitalWrite(ledPin, HIGH);
Output Serial Monitor
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM6')
New Line
115200 baud
[TERIMA CAHAYA] ADC: 62
[RULE ENGINE] Kondisi Kritis! LED Menyala (HIGH)
[TERIMA SUHU] 33.30 °C
[RULE ENGINE] Kondisi Kritis! LED Menyala (HIGH)
[TERIMA CAHAYA] ADC: 63
[RULE ENGINE] Kondisi Kritis! LED Menyala (HIGH)
[TERIMA SUHU] 33.30 °C
[RULE ENGINE] Kondisi Kritis! LED Menyala (HIGH)
[TERIMA CAHAYA] ADC: 63
[RULE ENGINE] Kondisi Kritis! LED Menyala (HIGH)
Ln 70, Col 2 NodeMCU 1.0 (ESP-12E Module) on COM6


NodeMCU 1.0 (ESP-12...
Search

Dalam tugas akhir modul ini, saya bersama kelompok merancang sistem pemantauan lingkungan dan kendali otomasi terdistribusi menggunakan jaringan Ad-Hoc Mesh Multi-Hop tanpa router, di mana peran setiap node dibagi menjadi tiga fungsi utama:

- Node 1 (Stasiun Lingkungan - Sensor DHT):
    - Node ini bertugas membaca suhu dan kelembapan riil menggunakan sensor DHT11/DHT22 yang terhubung di pin D4 (GPIO 2) setiap 3 detik.
    - Data tersebut dikemas ke dalam string JSON berformat ringkas (misal: {"tipe":"suhu_node","suhu":29.4,"kelembapan":65.0}) dan disiarkan ke seluruh jaringan mesh.
- Node 2 (Stasiun Cahaya - Sensor LDR):
    - Node ini membaca nilai ADC analog dari rangkaian pembagi tegangan sensor LDR yang terhubung ke pin A0 setiap 3 detik.
    - Data tersebut juga dikemas ke dalam format JSON (misal: {"tipe":"cahaya_node","adc":420}) dan disiarkan melalui jaringan mesh.
- Node 3 (Hub Kontrol Aktuator - Rule Engine):
    - Tugas pengiriman pesan dinonaktifkan (taskSendMessage dimatikan) karena node ini beroperasi secara pasif sebagai penerima dan pengambil keputusan.
    - Melalui fungsi callback penerimaan, Node 3 membedah string JSON yang masuk dari Node 1 dan Node 2 menggunakan pustaka ArduinoJson.
    - Menerapkan Rule Engine: LED yang terhubung di pin D6 (GPIO 12) akan menyala (HIGH) jika suhu dari Node 1 melebihi 31.0°C ATAU nilai ADC cahaya dari Node 2 berada di bawah 300 (kondisi gelap), dan akan dimatikan (LOW) jika kondisi kembali normal.